

The Tree of Reality: An Evolutionary Theory of Everything

The machine-verified state of the Generator programme — the proven trunk, the stated assumptions, and the predicted gaps

Author: Mark E. Mala (pen name of Ekram Alam) **Date:** September 2026 **Series:** the capstone of the Infinitography / Convergence Codex programme — a single-author, machine-verified research programme asking whether the structures of physics are generated from a minimal self-referential root. This paper supersedes the running claim-registers of the programme's earlier documents (in particular the May 2026 register of *Paper F v5.1*) wherever they disagree; §8 lists every superseded claim explicitly. The underlying papers keep their DOIs and their history. **Verification:** every mathematical claim in this paper is backed by a machine-checked proof in the Lean 4 proof assistant, in the public repository wonderben-code/convergence-codex, and §10 gives the exact branch, commit discipline, and the two commands that let a reader check any of it without trusting the author.

Abstract

This paper states a theory and publishes its complete current state under machine verification. The **Generator theory** — the *Tree of Reality* — proposes that the structures of physics are not axioms but descendants: a single minimal algebra, forced by the logic of self-reference, grows by one repeated canonical operation into the dimension and signature of spacetime, the gauge structure of the Standard Model, colour, chirality, three generations, and the machinery of quantum mechanics. Where the unification tradition seeks one framework to *contain* the known structures of physics, this theory derives them from one *ancestor*: it is a theory of everything in the sense that evolutionary biology is a theory of every species — unification by common descent. §1 states the theory plainly; the rest of the paper grades every claim in it.

The grading vocabulary is unforgiving. The theory's chain from nothing to everything has twenty-four links. **Six are GENUINE** — their headlines are machine-checked theorems, resting on nothing but the three standard axioms of Lean's mathematics library. **Thirteen are PARTIAL** — real theorems cover part of the headline, and the unproven residue is named precisely, in every row. **Three are POSTULATES** — modelling choices stated rather than hidden. **Two are OPEN** — and the paper's closing sections treat those two not as apologies but as its predictive content.

The prediction is this: the open gaps are exactly located, their missing mathematics is exactly shaped — down to the failing step of each staircase — and as machine reasoning in mathematics improves, the gaps will be filled and the tree shown whole. The prediction is dated, machine-checked at its boundaries, and anchored to the Bitcoin blockchain. Whoever fills a gap — human, machine, or the two together — completes the programme published here. A second class of prediction follows from the tree's branching: at four nodes the mathematics proves alternative branches exist, and structures generated on branches we do not inhabit are predicted to be undiscovered, not impossible — stated at its own evidential tier in §9.

The evidence discipline behind the labels is itself a result. Over a 45-working-day verification campaign (28 July – 26 September 2026, 243 numbered units), the corpus grew to **1,230 graded Lean files (312,334 lines), building green in 5,091 jobs, with exactly one sorry and one axiom in the entire build** — both predating the campaign, both documented, neither load-bearing for any claim below. In 169 consecutive units of hardening, **no link rating moved**: the work sharpened residues and closed named sub-gaps without inflating a single headline. Three of the programme’s own July headlines were *machine-refuted* by the campaign and are tagged down in this paper by the same hands that proved everything else. A theory that keeps its own score this way earns the right to call its gaps predictions.

1. The theory, in plain terms

This section states the theory on its own ground, for a reader meeting it for the first time. Nothing here is yet graded; every claim below is audited, link by link, in §4.

Physics, as normally written, begins with axioms: a spacetime of stated dimension, a gauge group chosen to match experiment, particle content and constants supplied by hand. The Generator theory proposes that these are not starting points but *descendants*. There is, it claims, a growth process — a small set of forced moves, iterated — that begins with nothing but the logic of self-reference and produces, level by level, the recognisable structures of physics. On this view the right question about any such structure is not *why this axiom?* but *which branch of the tree is this, and what forced it to grow?*

1.1 Ancestry, not unification

Every prior programme called a theory of everything has, at bottom, sought a **container**: one framework — one set of equations, one symmetry, one geometry — large enough to hold the known structures of physics side by side. The structures themselves are taken as given; the work is to reconcile them. This theory belongs to a different genus. It does not ask *what frame contains the branches of physics?* It asks *what did they descend from?*

The precedent is not in physics but in biology. Before common descent, the species were a catalogue — each one a separate fact, and every resemblance between them a curiosity. Darwin’s move was not to find a framework that contained the dog and the octopus; it was to show that they are *relatives*. You do not unify a dog and an octopus. You find their common ancestor — and then the dog, the octopus, and every species between them fall into one tree, and the tree explains not merely that they coexist but *why they share their deepest structures*.

The claim of this paper is that physics admits the same move. All of it — spacetime and its signature, the gauge forces, colour, the generations, the quantum apparatus — descends from a single ancestor: the minimal self-referential seed of §1.2, grown by one repeated operation. On this reading, the familiar resemblances across physics — the same small algebras surfacing in sector after sector — are not coincidences awaiting a container. They are **homologies**: the same bones in the whale’s flipper and the bat’s wing, inherited from the same ancestor.

This is still unification — arguably the strongest form of it. Common descent unified biology more completely than any single equation of life could have, because it explains kinship rather than mere

compatibility, and because it turns a catalogue into a history with a direction. The cascade's proven irreversibility (§1.2) gives this tree the same property: not just a family, but a family with an arrow.

Where the analogy is exact, and where it is not. Exact: a single ancestor; branching descent; an irreversible direction; homology as inherited structure; and a tree that is tested at its gaps — biology's tree was vindicated by fossils found where the tree said to dig, and §9 of this paper is, deliberately, a list of dig sites. Not exact: biological evolution runs on variation and selection — chance, filtered. The cascade has neither. Every step is *forced*: this is a derivation tree, not a selection tree, closer to the way a proof unfolds than to the way a population drifts. The analogy claims ancestry and homology; it does not claim natural selection, and nothing in this paper depends on it.

And the difference from the existing programmes, stated without adjudicating any of them: a container theory takes as input the structures it unifies, so it can be incomplete but is hard to refute *on structure*; an ancestry theory must **derive** its structures, so every branch is an exposure. If the tree grows a wrong branch — a gauge structure physics does not have, a signature it forbids — the theory fails there, visibly. The grading vocabulary of §2 exists precisely because ancestry claims are refutable in a way containers are not, and §8 shows the machinery catching this theory's own wrong branches. Nothing here argues against the unification programmes: ancestry answers a different question — not *what law is deepest?* but *what did the laws grow from?* — and a finished physics may well need both.

1.2 The tree, from root to crown

The root is a logical fact, not a physical one — and the tree genuinely begins at nothing. Its first node is $\emptyset$: not a vacuum, not a field, not a law waiting to act — nothing. In any setting where description can be turned on itself, fixed points are unavoidable — this is Lawvere's fixed-point theorem, the single mechanism underneath Cantor's diagonal argument and Gödel's incompleteness. The theory reads that unavoidability as the seed of existence: a world in which reference is possible cannot be empty, because self-reference forces something for reference to land on. (The theorem is machine-checked — from no axioms at all; whether the reading is right is graded with everything else.)

The seed is the smallest algebra that can hold a non-commutative fact. Ask for the minimal structure in which the order of operations matters — the signature property of the quantum world — and, over the complex numbers, the answer is unique: $M_2(\mathbb{C})$, the two-by-two complex matrices. Physicists know this object as the observable algebra of a single qubit, the smallest quantum system there is. The tree grows from it.

One operation drives all growth. From any stage, pass to End of it — the algebra of all linear transformations of that stage; informally, *the mathematics of studying the previous stage*. Applied to the seed it yields $M_4(\mathbb{C})$; applied again, $M_{16}(\mathbb{C})$; again, $M_{256}(\mathbb{C})$. The dimension squares at every step, and the process is provably irreversible — growth has a built-in direction. This iterated End is the *generator* of the theory's name, and the sequence $M_2 \rightarrow M_4 \rightarrow M_{16} \rightarrow M_{256} \rightarrow \dots$ is called *the cascade*.

Three lineages branch from the growth. Every algebra in the cascade supports three canonical kinds of structure, and the theory tracks each as a lineage of the tree: **End** — what things *are* (matter, spacetime, structure); **Aut** — how things *transform* (symmetry, and so forces); and the **inner product** $\langle \cdot, \cdot \rangle$ — how things are *known* (measurement, probability, quantum mechanics). That these three are the *only* canonical lineages is a modelling choice the paper declares openly rather than a theorem.

And the ancestor has many descendants — this paper walks one branch. A common ancestor with a single line of descent would be a chain, not a tree, and the programme’s published formalism says as much: its origin statement generates a *family* of self-referential fixed points — the chain $\emptyset \rightarrow \mathbf{I} \rightarrow \mathbf{I} \oplus \mathbf{I} \rightarrow \{\mathbf{D}^\infty \wedge \mathbf{C}\}$, one fixed point for every categorical setting $\mathbf{C}$ — with the known structures of physics identified as “*structural properties of specific lineages*” that “*share a common ancestor.*” The branch this paper walks — complex seed, Lorentzian signature, the Pati–Salam decomposition, three generations — is the one whose crown we inhabit. Where the walk meets a fork, the paper says so, and §9 returns to what the other branches mean.

The branches are the recognisable face of physics. At the cascade’s second level, $M_4(\mathbb{C})$ is exactly the Clifford algebra of four-dimensional spacetime — the algebra Dirac built the electron’s equation from — and it is that algebra *only* in dimension four: the theory’s account of why space and time have the dimensions they do. At the next level, M_{16} carries the Pati–Salam structure long identified as a natural home of the Standard Model; inside it, the fundamental four splits as $3 \oplus 1$ — three colours of quark beside one lepton. The quaternions — the last real division algebra Frobenius’s theorem allows — have exactly three independent imaginary directions: the theory’s candidate for why matter comes in three generations. A Higgs sector appears with its coupling bracketed inside a proven window. And the three lineages reunite in one object — the spectral triple: one algebra, one Hilbert space, one Dirac operator — on which a single functional, the spectral action, is claimed to produce both gravity and the gauge forces from one formula.

What kind of theory this is. It stakes structure first and numbers second: of physics’ measured constants, exactly two have so far been computed from the tree’s own objects — the weak-mixing ratio $3/8$ as a trace identity, and the Higgs coupling window — and the paper states this inventory up front, because a theory’s credibility begins at its inventory. Its claims live at deliberately different strengths: some links of the chain are machine-checked theorems, some are partly proven with the residue named, some are declared modelling choices, and two are open. The discipline of saying *which is which*, link by link, is the entire remainder of this paper.

1.3 The claims, in one place

For the reader who wants the theory’s full commitment on a single page, its claims are these, in ascending order of boldness:

1. **From nothing, something is forced.** The tree begins at $\emptyset$; self-reference makes emptiness untenable, a seed algebra follows, and one repeated operation grows the tree.
2. **The tree drawn in this paper is the branch of *known* physics.** What §4 renders is our universe’s physics as currently established — spacetime, gauge structure, colour, generations, the quantum apparatus — traced to the common ancestor and graded link by link, machine-checked wherever the tag says so.
3. **The gaps in our branch will close.** As machine reasoning in mathematics advances, the named gaps of §9 will be filled in the shapes specified, and the wood of the tree will harden — the first class of prediction.
4. **The seed has other descendants.** Alternative physics exists as coherent mathematics — proved, at four named forks — and is predicted to be *undiscovered rather than impossible*: possibly present in our own universe and not yet found, possibly realised beyond it (the latter at the plenitude

conjecture’s own evidential tier). Discovery of such a structure confirms a prediction dated here — the second class.

5. **Any newly found physics lands in one of three places.** It extends our branch, or it realises a sibling — both predicted — or it fits nowhere on the tree, and that third outcome refutes the generative claim itself. The theory names its own failure mode in advance.

Taken together, the five claims make the tree a **prediction instrument that points both ways**: outward, it says where to dig and what may exist — the gaps of our branch, and the structures of the branches beside it; inward, it says where anything newly found must land — filling a gap, joining our branch, or matching a sibling — with “nowhere” as the theory’s own named refutation. Everything after this point is the evidence, the grading, and the machinery that keeps these five claims honest.

2. The reader’s contract

This section fixes what every label in this paper means, so that no sentence below can say more than the mathematics does. Two working terms first: **the estate** is the programme’s complete formal corpus — every Lean proof file and ledger in the repositories of §10 — and **the campaign** is the 45-working-day verification effort (28 July – 26 September 2026) that graded it, working in 243 numbered **units**, each one sealed cycle of proof, audit and commit.

GENUINE. The link’s headline is a machine-checked theorem: no sorry, standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), and the formal statement says what the headline says. Per-file axiom audits (`#print axioms`) back each one.

PARTIAL. Real theorems cover part of the headline; the rest is stated as missing, in the link’s own row. The residue is always one of two kinds — open mathematics, or a postulate — and the row says which.

POSTULATE. The headline is a modelling choice. It is recorded in the assumptions ledger (§7), with what it bites and which way it cuts. A postulate is not a defect; an *unstated* postulate is.

OPEN. A precisely stated wall (§6) with no covering theorem: the staircase of steps toward it is climbed as far as it goes, and the exact failing step is named.

Two boundaries matter as much as the labels. **First**, a GENUINE link is a theorem about the estate’s mathematical objects; whether those objects are the physical ones is, link by link, a modelling question that the assumptions ledger answers separately. Machine verification removes doubt about the mathematics, not about the physics identification. **Second**, a wall’s failing step marks where the *known* routes stop — it is not a proof that no route exists. One of the campaign’s own walls fell by a route no document had listed.

Everything else in this paper is organised to be checked: §10 gives the repository, the branch, the stop-point commit, and the verification commands. The reader is never asked to trust; only to run.

3. The method, and why the labels can be believed

The tree was not labelled by opinion. It was labelled by a campaign with the following discipline, run from 28 July to 26 September 2026 in 243 numbered units.

Grading. Every Lean file in the estate carries a grade in a ledger (TRUE_LEDGER): GENUINE (1,142 files at the stop), HOLLOW (59 — files whose theorems compile but do not say what their prose claims; kept, labelled, and never cited as evidence), MIXED (15), PARTIAL (12), NOT_BUILT (2). The grade of a file is decided adversarially — by asking what its statements *fail* to say — and re-audited when touched.

Gates. Each unit passed a defect gate (44 modes, 25 scanners at the stop) before it could land. The gate's own vacuity census asks, of each law, whether it could pass while checking nothing.

Errata. The campaign filed **699 numbered errata** against itself and the estate — each one a sentence somebody wrote that the mathematics later contradicted, with the correction quoted beside it. The last correction was committed three minutes after the stand-down ledgers themselves — softening a claim in the campaign's own closing documents. The errata are part of the published record.

The stability result. Across units 75–243 — 169 units of continuous sharpening — **no spine rating moved**. Five separate recomputations re-read the whole table against the work and each found no rating change. The natural drift of a long project is upward relabelling; this campaign's drift was zero. That fact, more than any single theorem, is why the six GENUINE tags below can be taken at face value.

The refutations. The campaign proved theorems *against* three of the programme's own published July sentences (§8): the reflexive-domain headline over sets, the structural exclusion of Riemannian signatures, and the Standard-Model-in-su(4) assembly. Each refutation is itself machine-checked, and each is worth as much as a proof — a labelling system that can only move in the author's favour measures nothing.

4. The Tree, whole — the most complete form to date

This section is the paper. Everything before it is calibration; everything after it is support. What is drawn here is the **walked branch** — the physics of our universe as currently known, traced back to nothing — with the sibling branches of §9 implied at every fork the mathematics has proved. The figure below renders the Tree of Reality as of unit 243 — every link of the nothing→everything chain — the programme calls it *the spine* — carrying its audited tag, with the two open gaps drawn **in place**, marked ◇ and left deliberately visible.

The scoreboard: of 24 links — 6 GENUINE · 13 PARTIAL · 3 POSTULATE · 2 OPEN. (Read strictly three ways, the count is 6 GENUINE and 18 not; every PARTIAL row below names its residue, and the residue is always either open mathematics or a stated postulate.)

```

∅ nothing
└ L1 GENUINE    existence from self-reference (Lawvere's fixed point)
  └ L2 PARTIAL  the reflexive domain  $D \approx (D \rightarrow D)$ 
    └ L3 GENUINE the seed:  $M_2(\mathbb{C})$ , the unique minimal non-commutative algebra
      └ L4 GENUINE the cascade:  $M_2 \rightarrow M_4 \rightarrow M_{16} \rightarrow M_{256} \rightarrow \dots$  (End-iteration)
        └ L5 GENUINE irreversibility – the cascade cannot run backwards
          └ L6 PARTIAL why the fourth level – the CCM classification [gap W9 behind it]
            └ L7 POSTULATE three lineages branch:  $\text{End} \cdot \text{Aut} \cdot \langle \cdot, \cdot \rangle$ 
              └ THE END LINEAGE – matter, spacetime, gauge
                └ L8 GENUINE    spacetime dimension 4:  $\text{Cl}_4(\mathbb{C}) \cong M_4(\mathbb{C})$ , and only in dim 4
                  └ L9 PARTIAL  Lorentzian signature (1,3)
                    └ L10 PARTIAL the gauge algebra, embedded
                      └ L11 PARTIAL Pati-Salam structure at  $M_{16}$ 
                        └ L12 POSTULATE chirality – which factor is "left"
                          └ L13 GENUINE colour: the 4 splits as  $3 \oplus 1$ 
                            └ L14 POSTULATE three generations – the imaginary quaternions
                              └ L15 PARTIAL the Higgs sector and its coupling window
                                └ L16 PARTIAL anomaly cancellation on the chiral 16
                                  └ L17 PARTIAL the Weinberg angle:  $3/8$  as a trace ratio
                └ THE AUT LINEAGE – symmetry and gravity
                  └ L22  $\diamond$  OPEN    the Einstein equations from the spectral action [gap W5]
              └ THE  $\langle \cdot, \cdot \rangle$  LINEAGE – quantum mechanics
                └ L21 PARTIAL Hilbert space, GNS, Stone, Schrödinger
                  (the Born rule is a cited input, not a theorem – assumption 59)
              └ THE CO-EVOLVED CROWN – where the three lineages meet
                └ L18 PARTIAL the spectral triple (algebra, space, Dirac operator)
                  └ L19 PARTIAL the spectral action – the exponential cutoff forced
                    └ L20 PARTIAL the Boltzmann measure of the action
                      └ L23  $\diamond$  OPEN Yang-Mills and the mass gap [gaps W1–W4]
                        └ L24 PARTIAL the numbers: what has actually been computed

```

The two $\diamond$ boxes are the subject of §9. They are not embarrassments at the edge of the tree; they are its most precisely surveyed territory — each carries a climbed staircase and a named failing step, which is to say: a description of the missing piece.

The 24 links, each in one honest breath

What follows compresses the audited link table. For every link: what the headline says, what is machine-proved (with the proving files), and what is not — with the residue's kind. The full statement-level table, with every dated amendment, is in the published ledgers (§10).

L1 • Existence from self-reference — GENUINE. The root. Lawvere's fixed-point theorem at the programme's own domain (proved with *no* axioms at all — the file's axiom print is empty), with Cantor's theorem as its corollary, and the theorem re-proved in the cartesian closed category of ω -CPOs where the non-degenerate story lives. *Not proved:* the interpretive gloss (Gödel and Turing as consequences) — prose, and now labelled as prose. (*LawvereFixedPoint.lean* · *paper_f/DInfForces.lean*)

L2 • The reflexive domain $D \approx (D \rightarrow D)$ — PARTIAL. A non-trivial ordered structure satisfying $D \approx (D \rightarrow \text{continuous } D)$ exists — constructed, machine-checked. *Not proved:* the July headline read $D \approx (D \rightarrow D)$ with the **full** function space, and that statement is machine-refuted over sets: only one-point solutions exist. The continuous reading is the correct one, and adopting it as the published statement is an author's restatement, recorded as such. (*CanonicalTower.lean* · *ReflexiveDomainObstruction.lean*)

L3 • The seed $M_2(\mathbb{C})$ — GENUINE. Over $\mathbb{C}$, among finite-dimensional non-commutative algebras of the relevant class, dimension ≥ 4 is forced, dimension 4 is attained, and dimension 4 implies isomorphism

with $M_2(\mathbb{C})$; the $\ast$ -structure is carried along the isomorphism. *Scope stated:* over $\mathbb{R}$ the uniqueness is false (the quaternions also qualify) — the base field is a recorded modelling choice, and the Lean discloses it. (*SeedUniqueness.lean · paper_f/SeedStarStructure.lean*)

L4 · The cascade — GENUINE. The engine of the whole tree: $\text{End}(M_b) \cong M_{\{b \cdot b\}}$ at every size, the tensor isomorphism $M_n \otimes M_m \cong M_{\{n \cdot m\}}$, and the tower $\text{End}(D_k) \cong D_{\{k+1\}}$ proved at sizes $2 \rightarrow 4 \rightarrow 16 \rightarrow 256$, with a genuinely recursive construction equal to it level by level. *Not proved:* how deep the cascade runs, and which tensor factor decomposes — postulates, recorded. (*paper_f/F4_1a · CascadeEnd · SpineSharpenings · CascadeTowerRecursive*)

L5 · Irreversibility — GENUINE. Dimension strictly increases up the tower, and the level below is recoverable from End of it — the cascade is a one-way construction. *Labelled honestly:* “the arrow of time” is an interpretation; no theorem mentions entropy or time, and the files say so. (*paper_f/F4_1b · EmergenceLineage · SpineSharpenings*)

L6 · Why the fourth level — PARTIAL. Minimality of 4 under the stated criterion is proved with no upper bound assumed, and the first rung of the Chamseddine–Connes–Marcolli (CCM) classification is climbed: a finite-dimensional $\ast$ -algebra with a faithful $\ast$ -representation is a product of matrix algebras. On rung 2, the $\ast$ -structures are classified and the order-one condition is solved on every model bimodule the estate has. *Not proved:* the classification’s second half on CCM’s actual bimodule — the named failing step of gap **W9** — and the Standard Model algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ has no Lean declaration anywhere. (*CascadeMinimality · StarRepSemisimple · units 168–243*)

L7 · Three lineages — POSTULATE. That End, Aut and the inner product are *the only* canonical operations — the claim that makes the three branches exhaustive — has no formal statement; a fourth natural construction is already in use elsewhere in mathematics. Stated as the modelling choice it is. The lineages’ individual contents are credited to their own links.

L8 · Spacetime dimension 4 — GENUINE. The complex Clifford algebra in four dimensions is $M_4(\mathbb{C})$, for every nondegenerate form — and the converse: a complex Clifford algebra is $M_4(\mathbb{C})$ *only* in dimension four. The cascade’s fourth level and the algebra of spacetime are the same object, and dimension four is the only dimension at which that coincidence can occur. *Stated:* reading spacetime at cascade level D_2 is a recorded modelling choice. (*paper_f/CliffordIso · CliffordEvenLadder · SpineSharpenings*)

L9 · Lorentzian signature — PARTIAL. The real Clifford algebra of signature (1,3) is $M_2(\mathbb{H})$ on a form of proven signature; (2,2) is excluded structurally; the restricted Lorentz group and its double cover by $SL_2(\mathbb{C})$ are proved. *Machine-refuted:* the July claim that the Riemannian signatures (4,0) and (0,4) are excluded *structurally* — three signatures share one algebra, so Lorentzian-over-Riemannian is a postulate, now recorded as one. (*CliffordRealMinkowski · LorentzianChosen · SL2Connected*)

L10 · The gauge algebra — PARTIAL. Each Standard-Model factor embeds as a Lie algebra where the estate says it does, and the full Standard Model algebra embeds injectively in the Pati–Salam algebra. *Machine-refuted:* the July sentence “the SM algebra embeds in $\mathfrak{su}(4)$ ” for the estate’s own maps — the honest statement is the two-step one, and ~ 30 files’ prose is re-pointed accordingly (§8). The group level is a recorded postulate. (*SMLieHom · SMEembeddingHonest · SMInPatiSalam*)

L11 · Pati–Salam at M₁₆ — PARTIAL. The isomorphism $M_4 \otimes M_4 \cong M_{16}$ is genuine; Skolem–Noether and the automorphism structure are proved; the $\ast$ -structures on M_{16} are counted. *Not proved:* “(4,2,2) with no alternatives” — every factorisation $a \cdot b \cdot c = 16$ decomposes M_{16} and nothing in the mathematics prefers one; the choice is carried by a recorded constraint system. (*F1_6_PatiSalamForced · CascadeEnd · SkolemNoether*)

L12 · Chirality — POSTULATE. The asymmetry that makes the weak force left-handed is, in the estate, a definitional convention: the mirror statement compiles with the same one-line proof. This paper says so plainly — parity violation is *modelled*, not derived, and the row that once said otherwise is corrected in §8. (*supporting structure: TransposeSplit · Su2ModuleSixteen*)

L13 · Colour $4 \rightarrow 3 \oplus 1$ — GENUINE. On the cascade’s own carrier: the block embedding of $\mathfrak{gl}_3$, the invariant quark and lepton submodules, the singlet, $B-L$, and the triplet $\mathbb{C}^3 \cong$ the quark subspace intertwining the actions — irreducible inside the 4. *Stated:* identifying the carrier with the $SU(4)$ fundamental is a recorded assumption; no group-level statement. (*ColourEquivariance · ColourTriplet · ColourCascadeSubmodule*)

L14 · Three generations — POSTULATE. Frobenius’s theorem (the real division algebras are $\mathbb{R}, \mathbb{C}, \mathbb{H}$) is proved; the pure quaternions form a 3-dimensional space; the complexification identities hold at every size. The *identification* — those three directions ARE the three generations — is a physical postulate, recorded, and the mathematics itself proves the complexification cannot decide the real form, so the postulate is genuinely load-bearing. (*RealDivisionQuaternionCase and siblings*)

L15 · The Higgs sector — PARTIAL. The coupling window $g^2/N \leq \tilde{\lambda} \leq g^2$ with both endpoints sharp, the mass-squared window, and — new in this campaign — the Pati–Salam $\rightarrow$ Standard-Model breaking *classified:* unbroken groups, broken-generator counts, topology, for pairs of vacua at every rank-one first vacuum. *Not proved:* any scalar potential; the vacua are chosen, not derived; $N = 96$ is a recorded choice; no 125 GeV claim is made anywhere. (*HiggsBridge · units 199–227*)

L16 · Anomaly cancellation — PARTIAL. The full perturbative anomaly table vanishes on the chiral 16 — the sixteen fermion states of one Standard-Model generation in the Pati–Salam representation — evaluated on the actual representation in one theorem. *Cited, not proved:* that the cubic trace IS the triangle anomaly (ABJ), and Witten’s global anomaly (Mathlib has no homotopy groups). The fermion content is an input. (*AnomalyTraces · PatiSalamOnSixteen*)

L17 · The Weinberg angle — PARTIAL. In exact rational arithmetic on the chiral 16: $\text{Tr}(T_3 L^2)/\text{Tr}(Q^2) = 3/8$, and $\text{Tr}(T_3 L^2)/\text{Tr}(Y^2) = 3/5$ — with a control theorem showing the same construction on dimension-counting gives $3/7$, which *refutes* the programme’s old “ $3/8$ from the dimensions” mechanism (§8). *Assumed:* the coupling matching that turns the trace ratio into the physical angle at unification. (*WeinbergIndex*)

L18 · The spectral triple — PARTIAL. A complete finite real spectral triple of KO-dimension 6 exists for $M_n(\mathbb{C})$ on a four-block space, and the estate’s own triple on the regular bimodule of $M_2(\mathbb{C})$ is understood completely. *Not proved:* that it is the *cascade’s* triple — nothing connects it to the cascade’s Hilbert space; which real structure the cascade carries was an open author decision, ruled on 27 September 2026 (the Chamseddine–Connes KO-6 convention; §7). (*KOSixSpectralTriple · KOSixRealStructure · units 164–172*)

L19 · The spectral action: the exponential forced — PARTIAL. The genuinely striking half is proved: a factorising cutoff must be exponential (the Cauchy functional equation, proved in Mathlib — Lean’s community mathematics library — with no shortcuts), and on a tensor-sum operator the cutoff trace factorises. *Not proved:* the premise. No theorem says the cascade’s Dirac operator IS a tensor sum — the estate has no cascade Dirac operator at all — and the campaign’s final units settled exactly when the order-one condition supplies that shape: for one full matrix algebra at one generation, and *not* for products or generations. The “zero free parameters” headline of the May register does not survive this row (§8). (*F4_1h · SpectralCutoffFactorises · KronSumCriterion · ProductKronSum*)

L20 · The Boltzmann measure — PARTIAL. A genuine 16-dimensional Gaussian probability measure exists on the Hermitian slice, every polynomial moment finite. *Ruled (27 Sep 2026):* the Gaussian is the programme’s *stated approximation*; the true weight $e^{(-S)}$ exists in no Lean declaration, and the row says so. (*GaussianProductMeasure · Herm4Gaussian*)

L21 · Quantum outputs — PARTIAL. GNS on the trace state of the cascade algebra — positivity derived, cyclic vacuum, state recovery, faithfulness; Stone’s theorem in both directions for bounded generators in any C^* -algebra, instantiated on the cascade’s carriers; the Schrödinger equation at every finite level. *Cited, not derived:* the Born rule, Gleason, Wigner — physics inputs, recorded. *Bounded generators only:* the unbounded case is gap W2’s named step. (*CascadeGNS · FiniteStone · StoneConverseLocal*)

L22 · ◇ OPEN — the Einstein equations. The predicted element at gap W5. What exists: the algebraic Lovelock classification (every additive, homogeneous, equivariant map on curvature tensors is $\alpha \cdot \text{Ric} + \beta \cdot S \cdot \delta$ — proved), and the geometric chain to rung 3. What is missing, named exactly: the heat-kernel expansion of the spectral action — which requires unbounded self-adjoint operator theory that exists nowhere in Lean’s ecosystem. §9 states the prediction.

L23 · ◇ OPEN — Yang–Mills and the mass gap. The predicted element at gaps W1–W4. What exists: reflection positivity for the massive lattice field (measure level, every dimension), OS1 in finite volume, transfer-matrix gaps computed exactly for stand-in models — and a *measured refutation* showing the old uniform-bound target is false above the critical point, which is why the target was re-stated by ruling this September. What is missing, named exactly: the infinite-volume limit (no projective system exists to take it along), and an estimate that sees the transfer matrix’s structure below threshold. §9 states the prediction.

L24 · The numbers — PARTIAL. Exactly two physical numbers have been computed on actual objects: $\sin^2\theta_W = 3/8$ as a trace ratio, and the Higgs coupling window. Everything else — Newton’s constant, the cosmological constant, fermion masses, proton lifetime — is uncomputed, and this paper says so in one sentence rather than seventeen tables. What the May register claimed here is superseded (§8).

5. What fell — the campaign’s harvest

The tags above are the theory’s state. This section is what two months of machine verification added to it — the wood that thickened even while no headline moved.

Three walls closed outright. - **W6** (9 August): the Stein class and the smooth-compactly-supported Sobolev space $W^{\{1,2\}}(\gamma)$ are one class — in every dimension. A question the programme could not settle in July is now a biconditional. - **W7** (8 August): the real Clifford classification and the spin identification, with the full mod-8 periodicity table proved ten days later for every nondegenerate real form. - **W8** (12 August): a **non-trivial reflexive domain** — an ordered structure genuinely satisfying $D \models (D \rightarrow c D)$ — constructed and verified. The root of the tree grew its own witness: the object the whole programme is named for exists, machine-checked.

Lorentz and spin, proved properly. The restricted Lorentz group as the identity component of $O(1,3)$ with no hypothesis; the double cover $SL_2(\mathbb{C}) \rightarrow SO^+(1,3)$, surjective, kernel exactly $\{\pm 1\}$; the Clifford spin group modulo $\{\pm 1\}$ identified with $SL_2(\mathbb{C})$ modulo $\{\pm 1\}$; the complex Clifford classification in every rank.

Analysis that outgrew its scaffolding. The Gaussian Poincaré inequality in every dimension at every variance; Hermite completeness; Wick's theorem at every order for the finite-volume Gaussian field; Stone's theorem in both directions for bounded generators in any C^* -algebra; Frobenius's classification of the real division algebras; reflection positivity for the massive lattice field at the measure level — the step wall W1 was *named* for, which fell by a route no planning document had listed.

The Standard-Model side, computed on actual objects — not on numerology. The Weinberg ratio $3/8$ as a trace ratio on the chiral 16, in exact rational arithmetic, with a control theorem that kills the old dimension-counting story; the full perturbative anomaly table vanishing on the actual representation; the Standard Model algebra embedded injectively in the Pati–Salam algebra; and — the campaign's late crown — the **Pati–Salam $\rightarrow$ Standard-Model breaking classified**: the unbroken group at every relevant vacuum, the broken-generator counts, and the topology, for pairs of vacua at every rank-one first vacuum.

The seed understood completely. $M_2(\mathbb{C})$ as the unique minimal seed over $\mathbb{C}$; every $*$ -structure on $M_n(\mathbb{C})$ classified — $n/2 + 1$ classes up to conjugacy; a faithful $*$ -representation selecting the conjugate transpose.

The order-one chain, run to the end. A chain of units spanning 168–243 solving the order-one condition on every model bimodule the estate possesses — with the real structure, with the grading, with generations, on products — ending in an exact criterion: the tensor-sum shape that the spectral action's factorisation needs is **forced precisely for one full matrix algebra at one generation, and not otherwise**. That theorem is what disciplines link L19 and retires the May register's proudest headline (§8).

And five refutations, each a theorem. A non-trivial $D \models (D \rightarrow D)$ over sets — impossible. The Riemannian signatures excluded structurally — false; three signatures share one algebra. The Standard Model assembled inside $su(4)$ on the blocks used — false. $3/8$ from the dimensions — refuted by the $3/7$ control. The estate's own comparison route for wall W3 — proved insufficient, exactly ($O(n)$ against a target of $O(n^2)$). A programme that can prove theorems *against itself* is measuring something.

One more finding is a measurement rather than a theorem, and it matters below: wall W4's old uniform-bound target is **false above the critical point** — computed by diagonalising the estate's own transfer matrices, where the eigenvalue ratio climbs to 0.99990 at width 13. Two proof routes once recorded as

failures were not lossy; they were correctly reporting that the thing to be bounded does not exist. The target has been restated accordingly (§7).

6. The gap survey — six open walls, each with its failing step named

A gap is only a prediction if what fills it is described in advance. This section is that description: for each open gap — six walls behind the tree's two $\diamond$ links and one PARTIAL — the staircase that has been climbed, the exact step where the known routes stop, and **what would have to exist** for the gap to close. That last item is the predicted shape of the missing piece.

W1 · The lattice field's OS axioms — failing step: the infinite-volume limit. The finite-volume programme is done: reflection positivity proved at the measure level in every dimension, OS1 proved in finite volume. The limit is not a missing estimate — it is a missing *construction*: the finite-volume Gaussians are not a projective system (restricting a box's Green function does not give the smaller box's), so no Kolmogorov-style argument applies. **What would have to exist:** uniform-in-volume correlation bounds with a tightness argument, or an independent infinite-volume construction with a convergence proof. Genuinely open mathematics.

W2 · The continuum field and OS reconstruction — failing step 2b: an unbounded self-adjoint Hamiltonian. The Hilbert space and vacuum come from GNS, already consumed by the estate. The Hamiltonian that reconstruction produces is unbounded — and **Lean's mathematics library contains no unbounded operator theory at all**: no densely defined operators, no adjoints and closures, no spectral measure, no Borel functional calculus. Step 2b is not blocked on a theorem; *the objects the theorem is about do not exist yet in the formal ecosystem*. **What would have to exist:** the unbounded-operator subject itself — a library-scale project, not a research riddle.

W3 · Symmetry breaking (the Peierls argument) — failing step: the comparison bound. The combinatorial half is climbed: contours, cycle decompositions, the enclosure parity theorem. The estate's own comparison route is *proved insufficient* — its output is computed exactly, and it is $O(n)$ against a target of $O(n^2)$. **What would have to exist:** two recorded author decisions, then a comparison model of a genuinely different shape whose output is not sublinear. Elementary but voluminous; the gap is a theorem that the old method cannot close it, not evidence the statement is false.

W4 · The mass gap — failing step: a uniform spectral estimate below threshold. The one-dimensional chain is solved completely (whole spectrum, gap $2e^{-|\beta|}$, derived). The old uniform target was *measured false* above the critical point and has been restated by ruling to the high-temperature phase, where it is believed true and proved nowhere. **What would have to exist:** an estimate that sees the transfer matrix's structure — Dobrushin uniqueness or a high-temperature cluster expansion — neither of which exists in Mathlib. Genuinely open formalisation of genuinely standard physics.

W5 · The Einstein equations from the spectral action — failing step: rung 4, the heat-kernel expansion. Three rungs of differential geometry are climbed, including the algebraic Lovelock classification (every additive, homogeneous, equivariant candidate is $\alpha \cdot \text{Ric} + \beta \cdot S \cdot \delta$ — the *uniqueness* half of the Einstein equations, proved). Rung 4 needs the Dirac operator as an unbounded self-adjoint operator with discrete spectrum, its heat semigroup, the parametrix construction, and the coefficient

identification. **What would have to exist:** the same missing subject as W2, then a research-grade formalisation of the heat-kernel expansion. Two costs, and they should never be quoted as one.

W9 · The Chamseddine–Connes–Marcolli classification — failing step: rung 2’s second half on CCM’s own bimodule. Rung 1 is climbed (the campaign’s own theorem). Rung 2’s first half is climbed for complex factors. The order-one condition is solved on *every model bimodule the estate has* — the sixteen-unit chain of §5. What remains is the real thing: the isotypic decomposition of CCM’s H_F — a sub-bimodule of a product with three generations and a quaternionic factor — built *as an object*, with order-one computed across its pieces. One route is already closed by theorem (the doubled-algebra argument cannot run through a regular bimodule). **What would have to exist:** that decomposition, then K-theory of finite algebras for rung 3 — the latter absent from both the estate and Mathlib.

The one missing subject. Two of the six walls — W2 and W5, which are exactly the tree’s two $\diamond$ links — fail at the *same* absent mathematics: unbounded self-adjoint operators. One library, built once, unblocks both boxes. It is the largest single unlock the survey identifies.

And the honesty clause that governs this whole section: a failing step marks where the *known* routes stop. It is not a proof that no route exists. The campaign’s own record enforces the humility — W1’s named step fell by a route no document had listed, three weeks after the wall was declared.

7. What the theory rests on — the assumptions, stated

A theory’s honesty is measured at its inputs. The assumptions ledger records **57 numbered entries**: every choice the estate makes that no proof forced — a definition picked from several, a hypothesis nothing supplies, a physics input taken from the literature. An assumption is not a missing proof (those are §6’s walls); it can only be closed by changing the model or defending the choice.

Each entry carries a **bias grade**, and the grading is the ledger’s sharpest instrument: *anti-conservative* (the choice makes a result read stronger than it is — the dangerous direction), *conservative* (reads weaker), *neutral*, or *mixed*. The ledger grades **most of its own load-bearing entries anti-conservative** — it indicts its own choices where they flatter the theory. Its current states are equally blunt: *Stands*, *Narrowed*, *Hardenened* (later theorems showed the easy repair is impossible), *Corrected*, *Retired* (proved — three entries have graduated from assumption to theorem during the campaign).

The seven that carry the most weight, each in one line: 1. **The cascade record stores its own conclusions as fields** (entry 2) — the highest-leverage entry, invisible at its 21 consumption sites; a genuine recursive cascade now exists beside it, but no theorem yet marries the two. 2. **Which tensor factor decomposes** (entry 5) — parity violation rests on it; the mirror statement compiles with the same one-line proof. 3. **The trace state is the vacuum** (entry 16) — GNS’s headline results fail for a pure state. 4. **The spectral weight is multiplicative with unit moments** (entry 12) — the “zero free parameters” story dies exactly here (§8). 5. **The Higgs vacua are chosen, not derived** (entry 60). 6. **Three generations are the imaginary quaternions** (entry 25) — *Hardenened*: the mathematics itself now proves the complexification cannot decide the real form, so the postulate is genuinely load-bearing. 7. **The Born rule, Gleason and Wigner are cited physics, not theorems here** (entry 59).

Read against §1.2's family of descendants, the load-bearing postulates share a character worth naming: most are **indexical**. Which base field, which tensor factor, which signature, which vacuum — these do not shrink the tree; they state *which branch of it we inhabit*, the way a return address does not diminish a continent. The machine results of §9 make this reading precise rather than rhetorical: at each of those choices, the alternative is proved to exist.

One entry is the estate's only axiom (entry 53): a Gibbs measure with no defining property, predating the campaign, quarantined and disclosed — no campaign theorem depends on it.

Eleven entries are author decisions. The five highest-leverage decisions across the ledger and the campaign's running list were ruled by the author on 27 September 2026 — the rulings this paper is written under: the fermion content is a **named postulate** with derivation as an open target; the Gaussian is the **stated approximation** to the spectral-action measure, e^{-S} named as the true target; wall W4's target is **restated below threshold**; the real structure adopts the **Chamseddine–Connes KO-6 convention**; and the estate keeps **both spectral-triple notions under distinct names** — a strict CCM-compliant structure for anything this paper calls a real spectral triple, the weaker one as named scaffolding. Of the ledger's eleven decision entries this closes entry 48; the remaining ten stay open and are listed with the ledger.

8. The corrections — what this paper retires, in public

In May 2026 the programme's running register carried, in its own capitals: *"MASS GAP SOLVED, QG 100% COMPLETE, UNCONDITIONAL MILLENNIUM PRIZE PROGRAMME COMPLETE, ZERO FREE PARAMETERS."*

This paper supersedes that register, and this section does it explicitly, because a correction hidden in a footnote is not a correction. On 27 September 2026, twenty-six tag rulings — each proposed by the verification campaign with its reasoning, each re-checked five times against the estate — were accepted in full by the author. **No derived theorem was withdrawn by any of them.** The mathematics stands untouched; what moves is labels, in both directions, to match the mathematics exactly.

The loudest retirements: - **"Mass gap solved / QG 100% complete"** → the spine's honest tags are **OPEN** at L22 and L23. What exists are genuine theorems about the estate's internal models under stated hypotheses — real mathematics, wrongly headlined. The walls behind both links are §6's W1–W5. - **"Zero free parameters"** → **not established**. The claim's chain has an unproven premise: that the cascade's Dirac operator is a tensor sum. The campaign's final theorems settle exactly when order-one forces that shape — for one full matrix algebra at one generation, and *not* for the published algebra's actual structure — and what the conditions leave free at three generations is a whole matrix, not three moments. - **" $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ at the M_{256} level — PROVED ★"** → **META, OPEN**. The campaign's own digest calls this the largest gap between tag and reality it found: no Lean file constructs that algebra at that level; the files that name it do so to say what they do *not* prove. - **"The Weinberg angle $3/8$ from the dimensions"** → mechanism **refuted** by the $3/7$ control theorem; replaced by the stronger true statement — $3/8$ as a trace ratio on the actual representation, machine-checked in exact arithmetic. - **Three July headlines refuted outright** by the campaign's own theorems: the full-function-space reflexive domain (only one-point solutions over sets); the structural exclusion of Riemannian signatures

(three signatures share one algebra); the Standard-Model-in-su(4) assembly (not injective on the blocks used).

And five upgrades, by the same discipline: anomaly cancellation and Stone’s theorem rise from PREDICTED to PARTIAL; seed uniqueness from META-OPEN to PARTIAL; GNS to PROVED for the cascade algebra with the trace state, scoped; the emergence-lineage row to PROVED — a place where the paper had *undersold* its own Lean. A labelling system that can move in only one direction measures nothing; this one moves both ways and is therefore worth reading.

The full ruling-by-ruling record, with the campaign’s reasoning quoted, is published beside the ledgers (§10).

9. The predictions — the empty boxes, and what will fill them

A theory with gaps can hold them in one of two postures: apology, or prediction. This paper chooses prediction, and the logic of the choice is old and simple. When a structure is precise enough to specify its own missing pieces — where each gap sits, what shape the filler must take, what properties it must have — the gaps are not the structure’s weakness; they are its testable content. The canonical precedent is the periodic table, published with empty boxes whose described-in-advance occupants, once found, became the strongest evidence that the organising structure was real. The analogy claims a posture, not a stature: whether this tree earns any place in that history is exactly what the predictions below put at risk.

The Tree of Reality is published here in that posture. Its two $\diamond$ boxes and the wall behind its hardest PARTIAL are specified to the exact failing step: precisely where each box sits in the tree, which staircase reaches it, which step no known route climbs, and what the missing mathematics must look like when it arrives.

The prediction, stated once and plainly:

As machine reasoning in mathematics improves, each of the named gaps below will be closed, in the shapes specified — and the Tree of Reality will be shown whole. This prediction is dated, machine-checked at every boundary, and anchored to the Bitcoin blockchain. Whoever closes a gap — human, machine, or the two together — completes the programme published here.

The gap table

Gap	Location in the tree	The exact failing step	The predicted shape of the missing piece	Kind
G1	L23 ($\diamond$), wall W1	the infinite-volume limit of the lattice field	uniform-in-volume correlation bounds + tightness, or an independent infinite-volume construction with convergence	open mathematics

Gap	Location in the tree	The exact failing step	The predicted shape of the missing piece	Kind
G2	L23 (◇) and L22 (◇), walls W2 + W5	unbounded self-adjoint operators — absent from the entire formal ecosystem	densely defined operators, adjoints, closures, essential self-adjointness, spectral measure, semigroup $\leftrightarrow$ generator	library-scale infrastructure
G3	L23 (◇), wall W3	the Peierls comparison bound (the old route <i>proved</i> insufficient: $O(n)$ against $O(n^2)$)	a comparison model of genuinely different shape with non-sublinear output	voluminous combinatorics
G4	L23 (◇), wall W4	a uniform sub-top spectral estimate, below threshold (restated by ruling; the unrestricted form is measured false)	Dobrushin uniqueness or a high-temperature cluster expansion, formalised	open formalisation of standard physics
G5	L22 (◇), wall W5 rung 4	the heat-kernel expansion of the spectral action	the asymptotic expansion of $\text{Tr } f(D/\Lambda)$ with a_2 identified as a curvature integral — after G2 exists	research-grade, gated on G2
G6	L6, wall W9 rung 2	order-one across the isotypic pieces of CCM's actual bimodule	the isotypic decomposition of a finite-dimensional bimodule over products of $\mathbb{R}/\mathbb{C}/\mathbb{H}$ matrix algebras, built as an object; then K-theory of finite algebras for rung 3	construction + missing library

The six gaps are the headline predictions, but the claim extends down the whole tree: every PARTIAL link in §4 carries a residue that is named, stated, and machine-checkable, and the prediction covers those too. As the boxes fill and the residues close, **the wood of the tree hardens** — PARTIAL toward GENUINE, and each declared postulate either earns a derivation or remains, permanently and visibly, a stated input. The tree published here is the *most complete to date*; the prediction is that every future edition is more solid than this one, and that no honest edition will ever have to move a tag downward again — because this one already did that work.

Three structural facts sharpen the table. **First**, G2 fills two boxes at once — the tree's two ◇ links fail at the *same* absent subject. **Second**, the boxes are of different kinds, and the kinds have different closure profiles: library infrastructure (G2) is engineering — its arrival is a matter of effort, not discovery; combinatorics (G3) yields to patience; genuinely open mathematics (G1, G4) is the boldest part of the prediction, and we make it anyway. **Third**, G5 is *gated*: it cannot be attempted before G2 exists, which makes the predicted order of closure itself falsifiable content.

The sibling branches — the second class of prediction

Everything above concerns the branch we inhabit. The tree's own logic — §1.1's argument, and the published family $\{\mathbf{D}^\infty \wedge \mathbf{C}\}$ — says the ancestor has other descendants. This subsection states what that means, in three tiers whose evidential grades are deliberately different and deliberately labelled.

Tier 1 — machine-checked: the alternative branches exist as mathematics. The campaign proved, at four separate nodes of the tree, that the branch we walk is not the only coherent one — and these are the same theorems §8 reported as corrections, read now in their other direction: - **The seed forks over the reals.** Over $\mathbb{R}$, two seeds qualify — $M_2(\mathbb{R})$ and the quaternions $\mathbb{H}$ — proved non-isomorphic, with the same complexification: two distinct branches meeting every published constraint (RealDivisionQuaternionCase, realForm_not_determined_two). - **The sixteenth level forks.** Every factorisation $a \cdot b \cdot c = 16$ decomposes M_{16} , and the mathematics prefers none of them: Pati–Salam is our address there, not a theorem (scoreboard L11). - **The signature forks.** Three signatures share one Clifford algebra — the Riemannian worlds are not excluded (LorentzianChosen.three_signatures_one_algebra). - **The depth is not forced** (CascadeMinimality.not_forall_eq_four): the criterion that motivates four levels provably does not compel them.

One set of theorems, two honest readings: as corrections they discipline our branch; as geometry they draw the others.

Tier 2 — this paper's prediction: unfamiliar branches are undiscovered, not impossible. Walk the tree down a fork we did not take and the generator still generates: coherent algebraic worlds with quaternionic seeds, Riemannian signatures, other gauge decompositions. We predict that such tree-generated structures are **candidates for physics not yet found** — in our own universe (as undiscovered sectors, regimes, or realisations of the same mathematics) or beyond it (Tier 3). The practical face of this prediction: if the tree is ever said to “predict a physics we do not observe,” the correct reading is not *the tree is wrong* but *the tree has located a sibling* — with one strict exception, stated below, that keeps the claim honest. And the prediction cuts forward as well as back: any later discovery of such a tree-generated structure — in nature, in the laboratory, in the mathematics of another field, or in another realised instantiation — confirms a prediction that is dated, machine-specified and anchored *here*, before the discovery.

Tier 3 — the programme's published conjecture, at its own tier: the branches are realised. The programme's prior work — *The Infinite Ground* (2026), with the origin statement of *The Theory of Everything and the Origin of Reality* (2026) — argues from the convergence of five independent frameworks that all coherent configurations are realised: “*the space of actuality exhausts the space of possibility.*” Under that conjecture the sibling branches are not merely possible physics but actual ones — other instantiations, elsewhere in the ground. This paper inherits the claim at the evidential tier its source assigns it: a meta-empirical argument built on frameworks that are themselves interpretations, taken at face value — conjecture, clearly so, and published as such with its own DOI and date.

The trichotomy — what any newly discovered physics means for this theory. Stated as one claim, in advance: physics not yet known will, when found, do one of three things. **Either it extends the walked branch** — a new structure of our own lineage, joining the tree where the tree already has a place for it, strengthening it; **or it realises a sibling** — a structure from a branch we do not inhabit, confirming the second-class prediction above; **or it fits nowhere on the tree at all** — no branch of the generator, walked or unwalked, can grow it. The first two confirm the theory, and are predicted. The third refutes

the generative claim itself, and we say so now: a genuinely tree-orphaned physics is this theory's deepest possible failure mode, named before any candidate exists.

The boundary that keeps all of this falsifiable. A shield that deflects every failure would make the theory unfalsifiable, so the boundary is drawn here, in advance: **the tree is refutable on the branch it walks, and never excused by the branches it merely permits.** The walked branch is pre-registered — §4 fixes it, link by link, tag by tag, before any dispute — so a failure there (a walked-branch structure contradicted by observed physics, or a gap of §9 proved unfillable as restated) counts fully against the theory and may never be reclassified, after the fact, as a sibling. Conversely, the absence from observed physics of a structure the tree generates on an *unwalked* branch refutes nothing: that is exactly what Tier 2 predicts. Pre-registration is what makes the distinction principled rather than convenient — the address was written down before the mail arrived.

Why we predict closure — the trajectory argument

The evidence is the campaign this paper closes. In forty-five working days, one researcher and one reasoning machine produced 1,230 graded proof files; closed three walls outright — including W8, a construction that had stood open since the programme's walls were first named; proved the named step of a fourth wall by a route no planning document contained; and refuted three of their own published headlines. Every one of those events was, five years ago, out of reach of any machine on earth.

The instrument is improving faster than any instrument in the history of mathematics. We do not claim a calendar; we claim a direction. Formal libraries compound — every object built for one gap (G2's operators, G6's decomposition) permanently lowers the cost of every later attempt. The gaps named here are, by construction, the *best-specified open problems in the formal ecosystem*: each arrives with its staircase pre-climbed, its dead routes marked by theorems rather than folklore, and its target statement already stated in a machine-checkable idiom. They are, deliberately, the problems it is easiest for a rising capability to close.

What would count against us

Symmetry demands this list, and honesty enjoys it. The boundary of the sibling-branches subsection binds throughout: everything below concerns the walked, pre-registered branch, where failures count in full and no sibling excuse is available. The prediction fails in kind, not just in schedule, if: the restated W4 target is proved false *below* threshold too; the infinite-volume limit of G1 is shown not to exist for the massive lattice field; the isotypic computation of G6 yields an algebra *other than* the Chamseddine–Connes–Marcolli one; or the heat-kernel coefficient of G5, once computable, fails to contain the Einstein term. Any of these would be a machine-checked refutation of this paper's central claim, and we commit in advance to publishing it under this paper's own DOI lineage with the same prominence as a confirmation. The label discipline of §§2–8 is the reader's evidence that the commitment is real.

And one caution we owe the reader in the other direction: a failing step marks where the known routes stop, not where all routes stop. W1's own named step fell to an unlisted route mid-campaign. The boxes may fill sooner, and by stranger routes, than the table suggests.

10. Provenance and verification — trust nothing, run everything

The repository. All Lean sources, ledgers, and this paper: github.com/wonderben-code/convergence-codex (public). The verification campaign’s working registers — the graded file ledger, the spine table, the walls ledger with every dated amendment, the 57-entry assumptions ledger, the 699 errata, the 1,846-entry progress log — are deposited in the archival package published with this paper’s DOI; the five stop-point ledgers that summarise them, and the author’s rulings, are archived beside this paper in the public repository (papers/capstone2026-work/), and originate on the campaign branch below. Where any summary in this paper differs from a register, **the register is right and this paper is wrong** — and the correction belongs in the errata.

The stop point. Branch `claude/infinetography-formalisation-fbqxf3`; the snapshot stop point is the ledger commit following unit 243 (26 September 2026), with the five stop-point ledgers under `lean_verify/paper_f/ledgers/`. The author’s rulings of 27 September 2026, under which this paper is written, are archived beside this paper (`RULINGS_2026-09-27.md`).

The anchoring. Every commit in both repositories is Bitcoin-timestamped on a six-hourly schedule via OpenTimestamps. The programme’s twenty-seven prior papers carry dated Zenodo DOIs; this paper receives its own. The May 2026 predictions register of the earlier convergence-capstone lineage remains part of the stamped record under its own provenance; its empirical predictions are a separate instrument from the gap-predictions of §9 and are not restated here.

The two commands. To check the mathematics, no trust in the author is required:

```
lake build          # the estate: 5,091 jobs, green; one sorry + one axiom, both
                   # pre-campaign, both documented, neither load-bearing
#print axioms <thm> # any theorem: the campaign's files answer with the standard
                   # three axioms — propext, Classical.choice, Quot.sound
```

The per-file grades, the per-unit gate records, and `check_ledger.py --spine` (which recomputes the scoreboard from the link table and fails if they disagree) are in the registers.

11. Coda — publishing the table with its gaps

Here, then, is the tree as of today. Six links of solid, machine-checked wood — existence forced from self-reference, the seed, the cascade and its arrow, the algebra of spacetime, the splitting of colour. Thirteen links partly grown, each with its unproven remainder named in public. Three grafts declared as grafts. And two empty boxes, drawn in their exact places, with the properties of their missing pieces written beside them.

We could have waited to publish until the boxes were filled. We publish now, deliberately: a structure that can describe its own missing pieces is most useful *before* they are found, because the description tells whoever arrives what they are looking at. The boxes are offered in that spirit — to whoever, or whatever, arrives with the mathematics to fill them. The tree is planted, dated, and anchored. The gaps are the invitation.

12. References — the named results

Classical results are named where they are used; their primary sources are collected here. The paper’s real bibliography is the repository of §10: the programme’s twenty-seven prior papers (each with a dated DOI) and every Lean source this paper grades.

- F. W. Lawvere, *Diagonal arguments and cartesian closed categories* (1969).
- G. Frobenius, on real finite-dimensional division algebras: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ (1878).
- M. H. Stone, on one-parameter unitary groups (1932).
- E. P. Wigner, on symmetry representations (1931); A. M. Gleason, on measures on the closed subspaces of a Hilbert space (1957) — cited in §4 (L21) as physics inputs.
- J. C. Pati and A. Salam, *Lepton number as the fourth “color”* (1974).
- K. Osterwalder and R. Schrader, *Axioms for Euclidean Green’s functions* (1973; 1975).
- M. E. Mala, *The Infinite Ground* (2026), doi:10.5281/zenodo.19479968; *The Theory of Everything and the Origin of Reality* (2026), doi:10.5281/zenodo.19550042 — the possibility-space lineage inherited, at its own tier, by §9’s third class of claim.
- A. Connes, *Noncommutative Geometry* (1994).
- A. H. Chamseddine and A. Connes, *The spectral action principle* (1997).
- A. H. Chamseddine, A. Connes and M. Marcolli, *Gravity and the standard model with neutrino mixing* (2007).
- L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language* (2021); The mathlib Community, *The Lean mathematical library* (2020).

Mark E. Mala, September 2026. The author thanks the reasoning machines this programme was built with — including the one that spent forty-five days proving, refuting, and grading the claims above, filed 699 corrections against the work and its own, and whose final act before standing down was to soften a claim.